
Mitigating Misclassifications from Adversarial Attacks via a Hybrid Approach

Using Variant Classifiers to Determine Whether Images Are Created Using AI
Models

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Abstract

In recent years, classification models have shown promising results by taking a wide swath of data and properly classifying them with a high degree of accuracy in a wide range of contexts. However, the attack surface of these classification models has exploded due to the emergence of adversarial examples. These are well-crafted examples that hoodwink the model to produce confident but incorrect outputs. Thus, the need to suppress the effect of adversarial attacks on these otherwise robust classification models remains an ongoing field of research that we wish to explore. We proposed a hybrid approach to dealing with adversarial attacks by combining 2 different GAN classification methods: GAN Fingerprints and Defense-GAN. While Defense-GAN reconstructs adversarially attacked images, our results show that the reconstructed "clean" image leads to reduced attribution performance compared to the GAN Fingerprint classifier alone.

Code: https://github.com/smarte376/MassDataMining_project

1 Introduction

1.1 Motivation

The past few years have seen an explosion in the AI sector, particularly in the area of Generative AI (GenAI). These AI advancements have made huge strides in two categories: text-based Large

Language Models (LLMs), which are capable of generating text in a conversational fashion (e.g. ChatGPT), and image-based networks that aim to generate photorealistic images or videos (e.g. Nano Banana). Because of the new-found capabilities of GenAI, there is now a vested interest to develop computational methods to differentiate between AI-generated content and human-made content as human inspection alone is becoming less and less reliable. A recent publication from February 2025 reported a success rate of 68.7% for judging between AI-generated and human-written text [10]. Our work falls into this body of research, as we explore methods to distinguish between real and AI-generated images based on consistent artifacts, known as fingerprints, left behind in images generated with an AI model.

1.2 Goal

Our goal is to build a robust classifier using a classifier variant called GAN fingerprints that can accurately identify whether an image was generated by a Generative Adversarial Network (GAN) or was real [6]. In the event an image is determined to be GAN generated, we then try to identify the source model. Because perturbations are known to hamper the ability to perform accurate model attribution, we strive to make it robust against adversarial attacks using a defensive method called Defense-GAN. The methodology of Defense-GAN is to filter out perturbations in adversarial attack samples and attempt to produce a clean image sample from the modified one. This clean image is then fed to the GAN-fingerprint classifier for classification. Originally in our proposal, we also wanted to see if an adversarial training-based GAN (AT-GAN) can be used in conjunction with Defense-GAN's defense method to better filter our perturbations in adversarially attacked images before feeding the result to the GAN-fingerprint classifier. Unfortunately, we ran into several technical issues in setting up GAN-fingerprint to run with Defense-GAN and did not have sufficient time to pursue this idea further [4][3].

1.3 Main Objective

The main objective of this project is to create a robust classifier that can accurately classify GAN-generated images, even when under some form of adversarial attack. We consider a test-time adversary that applies image-space perturbations without access to the attribution model's internal parameters. This allows us to focus on model robustness under realistic scenarios where a would-be attacker would likely add noise, compress, or crop an image to evade proper classification. By reconstructing a cleaned version of an image that had undergone adversarial perturbations with Defense-GAN, the fingerprints that a generative model leaves behind during image generation would be sufficiently restored to enable the GAN-Fingerprint classifier to classify with comparable accuracy to that of an

untouched sample. Our hybrid approach would render such attacks ineffective, thus strengthening the reliability of model classification.

2 Data Sets

For this project, we plan to use 2 different but related data sets to enrich our study on GANs: CelebA human face data set containing over 200,000 facial images RGB with 10,177 unique identities and we select 200,000 RGB images from the LSUN bedroom scene data set. Both data sets are rich in the number of annotations, and they have the added bonus of being publicly available for download for the scope of educational use. Both data sets are well-suited for the purposes of this project because they see widespread use in recent literature regarding GANs, so there also exist adversarial data sets to ease the process of exploring defense mechanisms against adversarial examples and how robust they are to different types of modification. For our purposes, we aligned and cropped the CelebA data set at $(x, y) = (89, 121)$ with size 128x128. For the LSUN bedroom images we center-crop them to square size according to the shorter side length, and resize them to 128x128.

3 Our Approach

In our approach, we plan to create a GAN fingerprints classifier to explicitly visualize GAN fingerprints in the image domain in conjunction with Defense-GAN, a defensive method against adversarial attacks that attempts to remove perturbation noise in attacked samples and create a clean image. On its own a classifier that explicitly visualizes GAN fingerprints in the image domain has very high accuracy classifying clean images but struggles with certain adversarial attacks and combinations of them such as compression, cropping. Using them together, we hypothesized that it could potentially solve the accuracy problem against adversarial attacks, whereas Yu, Ning, et al fine-tune the classifier under the assumption that the classifier knows the attack category to immunize against attacks [6]. We tested this model against several GAN architectures such as ProGAN, SNGAN, MMDGAN and CramerGAN. For each model, we created 2,500 images per dataset (20,000 images in total) and then made test sets by randomly sampling images from all models and real images. In order to ease visualization and respect our space constraints, we sampled 500 from each GAN and 500 from the real images. The test sets were either clean (no attacks applied) or had at least 1 of 5 different attacks (noise, blur, cropping, JPEG compression, relighting) following Ning Yu, et al's procedure for applying attacks. Below are the adversarial attacks we applied to samples:

1. Noise: Apply Gaussian variance of $U[5.0, 20.0]$ at a per-pixel basis.
2. Blur: Apply Gaussian filtering with a random kernel size chosen from one of $\{1, 3, 5, 7, 9\}$.
3. Cropping: Apply random offset between 5% and 20% and then resize to the original size.

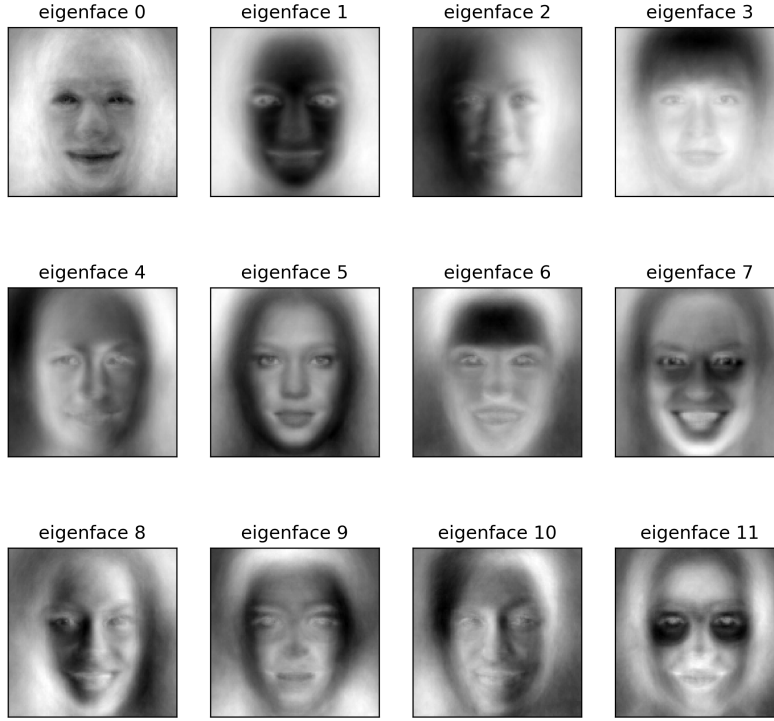


Figure 1: Top 12 Eigenfaces from CelebA training dataset

4. JPEG Compression: Apply lossy compression with quality factor randomly sampled from $U[10.0, 75.0]$.
5. Relighting: Apply SfSNet to replace the current image lighting condition with another random one from their lighting dataset [9].
6. Random Combination: Apply random mix of previous attacks with 50% probability for each [6].

We then implemented our own "baseline" classifiers that would attempt to classify images' source model based on different methods/features than their GAN fingerprints. For these baseline classifiers, we implemented a KNN approach over the raw pixels of the image as well as a SVM classifier over the top 1000 eigenfaces (PCA transformation of facial image data). Our top 12 eigenfaces are shown in Figure 1. Both classifiers were trained over only the clean data.

4 Evaluation

We assessed whether our method works by measuring its classification accuracy against 3 baseline classification methods: using the GAN-fingerprint's classifier on its own, k-nearest-neighbor (kNN) on raw pixels, Eigenface for the celebA test sets only [7]. We then used the following metrics for

comparison: model accuracy, f-1 scores, and AUC ROC score to evaluate how robust these methods are. We trained our own models for Eigenface and k-nearest-neighbors with the help of scikit-learn imports. Since Eigenface is meant exclusively for face recognition scenarios, we excluded it from any test sets for LSUN bedroom images. All baseline classifiers were trained exclusively on clean images, allowing us to isolate the effects of adversarial perturbations. The data from our results are given in the tables and figures below:

CelebA Data Set

Attack Type	K Nearest Neighbors	EigenFace	Fingerprint Classifier	Our Hybrid Approach
Clean	0.81, , 0.81	0.55, , 0.60	0.989, 0.9998, 0.989	0.16, 0.56, 0.24
Noise	0.28, , 0.28	0.41, , 0.44	0.73, 0.95, 0.75	0.16, 0.55, 0.24
Blur	0.27, , 0.27	0.24, , 0.28	0.927, 0.997, 0.926	0.19, 0.56, 0.25
JPEG Compression	0.28, , 0.28	0.29, , 0.32	0.36, 0.73, 0.40	0.18, 0.56, 0.25
Relight	0.25, , 0.25	0.41, , 0.44	0.91, 0.992, 0.91	0.17, 0.56, 0.24
Cropping	0.22, , 0.23	0.16, , 0.21	0.42, 0.84, 0.46	0.19, 0.56, 0.27
Random Combination	0.24, , 0.24	0.23, , 0.26	0.42, 0.73, 0.42	0.18, 0.55, 0.25

Table 1: Accuracy metrics for the classification methods on CelebA dataset. Eigenface and K-nearest neighbors use only f1-score and accuracy, Fingerprint classifier and our approach use f1-score, AUC ROC, and acc respectively

LSUN Bedroom Data Set

Attack Type	K Nearest Neighbors	EigenFace	Fingerprint Classifier	Our Hybrid Approach
Clean	0.85, , 0.58	n/a	0.94, 0.9991, 0.94	0.15, 0.52, 0.20
Noise	0.40, , 0.42	n/a	0.73, 0.93, 0.72	0.14, 0.52, 0.20
Blur	0.39, , 0.42	n/a	0.94, 0.997, 0.94	0.15, 0.52, 0.20
JPEG Compression	0.39, , 0.41	n/a	0.49, 0.81, 0.49	0.15, 0.52, 0.20
Relight	0.38, , 0.40	n/a	0.84, 0.98, 0.84	0.14, 0.51, 0.18
Cropping	0.33, , 0.33	n/a	0.70, 0.94, 0.70	0.15, 0.53, 0.20
Random Combination	0.34, , 0.35	n/a	0.48, 0.78, 0.47	0.13, 0.51, 0.19

Table 2: Accuracy metrics for the classification methods on LSUN bedroom dataset. K-nearest neighbors use only f1-score and accuracy, Eigenface does not work on bedroom dataset since it specifically is trained on facial images. Fingerprint classifier and our approach use f1-score, AUC ROC, and acc respectively

5 Discussion

Based off the results in tables and the ROC curves shown in Figure 4, it is clear to see that our hybrid approach is not only significantly outperformed by the standalone GAN Fingerprint classifier, it also has a ROC AUC score near 0.5. This tells us that our approach has a near random discrimination and

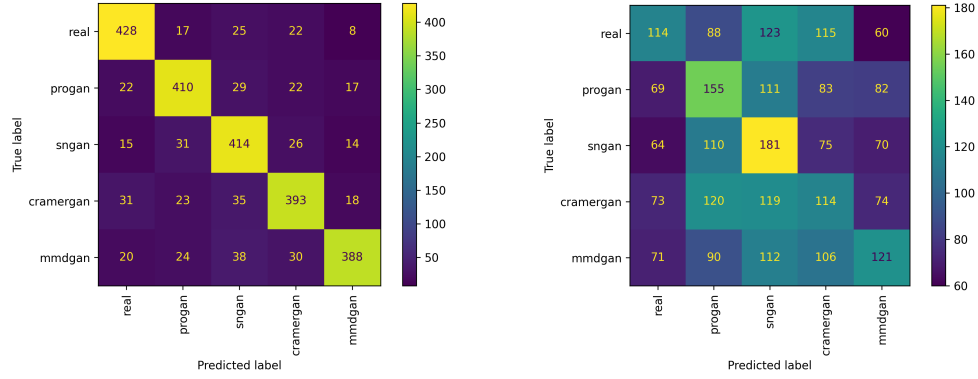


Figure 2: Sample confusion matrices for KNN on raw image pixels for clean images (left) and under blurring perturbation (right)

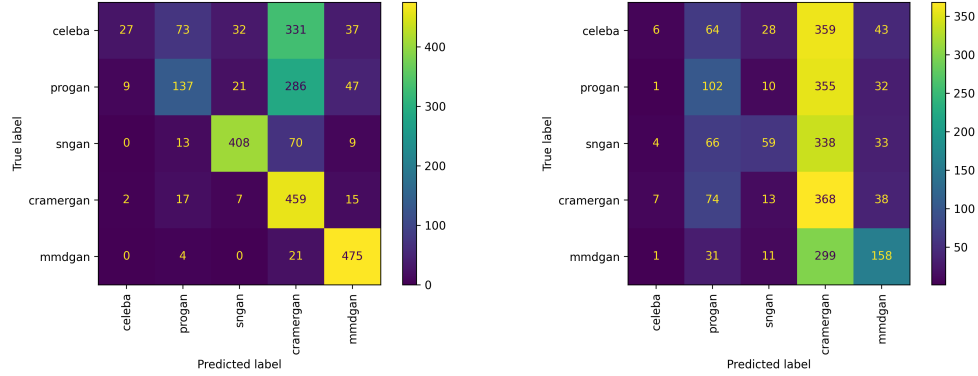


Figure 3: Sample confusion matrices for eigenface for clean images (left) and under blurring perturbation (right)

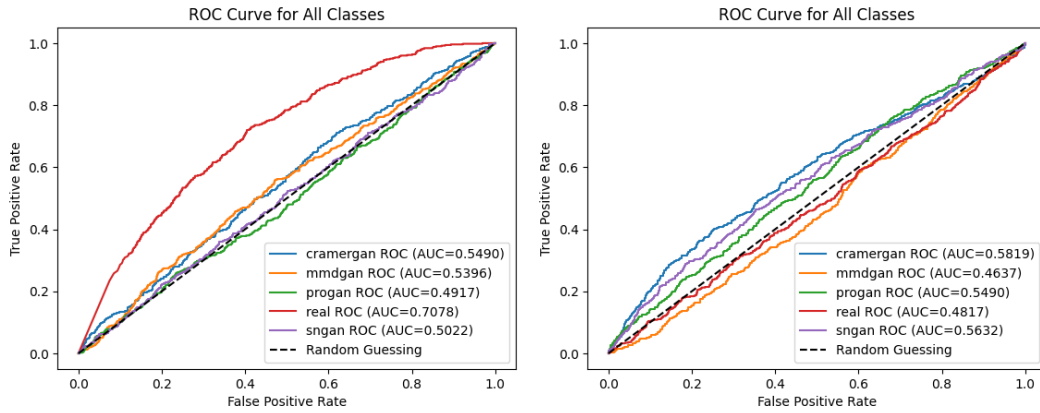


Figure 4: Sample ROC curves for our hybrid approach for images under cropping perturbation for CelebA (left) and LSUN (right)

cannot tell the models apart. The performance of the GAN Fingerprint classifier also outperforms K-nearest-neighbors and Eigenface which follows Yu et al's results but again our hybrid approach is outperformed by even K-nearest-neighbors. In our limitations section, we theorize a few reasons as to why our model performed so poorly but we believe the main reason is that GAN fingerprint's classifier which relies on the distinct model fingerprints GANs leave in their generated images in which Defense-GAN's reconstruction may be removing or disturbing. Supporting this hypothesis is the incredible consistency of our metrics across the different perturbation sets, with all metrics across all perturbations within a dataset being ± 0.3 of each other, meaning that the classifier had a similar performance on all DefenseGAN-reconstructed images regardless of the perturbation. This doesn't necessarily mean that DefenseGAN was successful in defending against said perturbations, but it does suggest that our model's performance was relatively independent of which attack was used.

Some other notable patterns emerge from the data. For example, our KNN implementation does a remarkably good job of classifying clean data, but is severely and consistently confounded by any sort of adversarial perturbation. This fidelity comes at the cost of runtime, with KNN runtimes being on the order of a few minutes during our testing, while our other classifiers had runtimes on the order of a few seconds or less on the same data. Our eigenface classifier, on the other hand, seemed to struggle to classify images very accurately, seeming to default to classifying images as being generated by the CramerGAN model, as can be seen in Figure 3. Eigenface also had more variance in classification accuracy, struggling the most with perturbations like blurring and cropping, whereas noise and relighting attacks performed better. As shown in Figure 1, the extracted eigenfaces recapitulated important "facial feature" information quite well. However, since the classification task at hand was recognizing GANs rather than identifying faces, it may be that eigenface did not extract information that was suitable for this task and instead lost important information (since it relied first on a PCA dimensionality reduction).

6 Limitations

While working on our project we came across a few limitations that impact the final results of our hybrid approach. One limitation was found when training our own model for defense GAN, Cifar10_A, when training was finished some metric data was given which told us that the average test loss of the model was 1.0743 and had an overall accuracy of 73%. Another limitation was the quality of the reconstructed images. When comparing a reconstructed image to its original counterpart, we found that the quality of the reconstructed image was significantly worse compared to the original (See Figure 5). We also hypothesized that in the reconstruction of an image, there is some (or complete) loss of fingerprint data in the image that the classifier uses to determine if the image is real or generated. Inspecting the individual results files ('results.csv' and 'metrics.txt' files

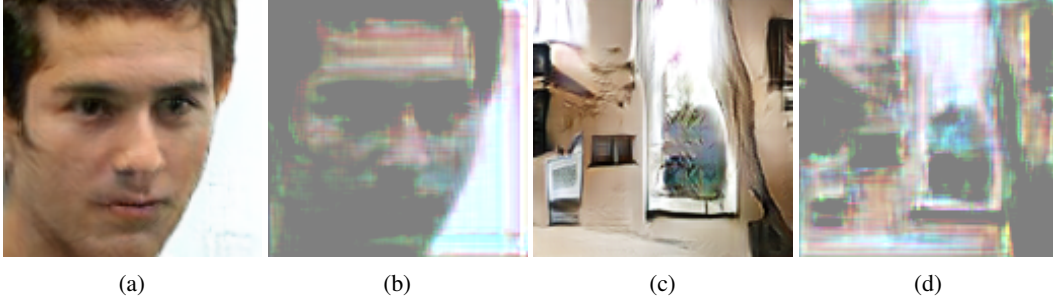


Figure 5: Original CramerGAN-generated images (a, c) with their DefenseGAN-generated counterparts (b, d). Left images taken from CelebA dataset under cropping perturbation; right images taken from LSUN dataset under random perturbation

included in our repo) seems to support this hypothesis, as the predicted class for most images was overwhelmingly "real" regardless of the true class. This would seem to suggest that the fingerprints the classifier was trained to search for were no longer preserved by the DefenseGAN transformation. We believe, the combination of these limitations accounts for a significant portion of the performance degradation compared to the GAN fingerprint classifier by itself. Another possible limitation is in how the fingerprint classifier was trained, it was not trained on reconstructed images which, depending on how much fingerprint data loss occurs during reconstruction, could have improved the performance of our hybrid model.

7 Related Works

Some relevant works are those by Yu, Ning, et al. "Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints" who introduce GAN fingerprints as an alternative classifier variant to explicitly visualize GAN fingerprints in the image domain [6]. Samangouei, Pouya, et al. "Defense-GAN: Protecting Classifiers against Adversarial Attacks Using Generative Models" propose Defense-GAN, a framework that uses the expressive capability of generative models to defend against adversarial attacks by modeling the distribution of unperturbed images to find a close output to a given image that does not contain adversarial changes, and the output is fed to the classifier. This method also has the added benefit of being able to be used with any classification model [3]. Lastly, Wang, Desheng, et al. "ATGAN: Adversarial Training-Based GAN for Improving Adversarial Robustness Generalization on Image Classification" proposes Adversarially Trained GAN (ATGAN) which adopts an image-to-image generator to map adversarial examples back onto the manifold of clean examples which helps to reduce the power of adversarial examples and also has an effect of data augmentation which increases the sample complexity of adversarial examples. ATGAN also incorporates adversarial training into standard GAN training to remove obfuscated gradients that lead to false sense in defending against adversarial perturbations [4].

Our idea tries to incorporate these different techniques to keep the high accuracy rating of non-adversarial attack images with the GAN fingerprint classifier and boost the accuracy of adversarial attacks with the help of a method similar to ATGAN/Defense-GAN.

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